104.58.1. Urban Metabolism and Flux Audit

1. Definition of Urban Metabolism

Urban metabolism describes the process through which a city:

Input
receives energy, water, food, and raw materials.

Internal processing
utilizes these resources through housing, industry, and transportation systems.

Output
releases waste and emissions into the surrounding environment.

EFU Principle

A city is sustainable when its **internal cycles** (circular economy mechanisms) maximize **value generation** while minimizing **environmental entropy**.

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2. Critical Flux Indicators (Urban EFU Metrics)

2.1 Import Dependency Ratio (IFR)

Measures the proportion of **external resources required for the survival of the city**.

Audit target

IFR < 0.4

A city operating in critical conditions should maintain **less than 40% external dependency**.

Interpretation

Low IFR indicates a **sovereign city** supported by local energy, water, and food systems.

2.2 Local Accumulation Capacity (LAC)

Measures the ability of the city to **temporarily store essential fluxes**, including:

• energy storage systems
• water reservoirs
• food reserves

EFU rule

Every urban district must ensure at least **72 hours of autonomous biological homeostasis** (drinking water and basic heating) in the event of technological disruption.

2.3 Entropy Emission Index (EEI)

Represents the degree of **thermodynamic and material dissipation** generated by the city, including:

• waste heat
• physical waste
• informational noise

Metric

EEI = Waste Output / Material Input

Higher EEI values indicate **lower metabolic efficiency** within the urban system.

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3. Types of Urban Fluxes

Flux Type	Source (Input)	Internal Transformation	Loss (Output)
Hydrological	precipitation, rivers, wells	purification, consumption	wastewater, evaporation
Energetic	solar energy, electrical grid, wind	heating, lighting, manufacturing	waste heat, light pollution
Biological	food supply, green areas	metabolism, recreation	organic waste
Informational	data networks, knowledge	decision-making, governance	digital noise, spam

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4. Audit Protocol: The “Urban Laboratory”

The EFU license requires **real-time monitoring of urban fluxes** through distributed sensor networks.

According to the principles defined in **Module 104.54 (Edge Computing)**:

• data processing should occur **locally**
• operational decisions (such as water pressure optimization or traffic control) must serve **local community interests**, rather than global profit-driven algorithms.

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5. Functional Outcomes of Module 104.58.1

This module enables the following capabilities:

• identification of **energy and resource leakage points** within the city
• detection of systemic **parasitic dynamics**
• establishment of a diagnostic foundation for the next stage of analysis

Next Module

104.58.2 — The Sovereign District and Autonomous Infrastructure

This module will examine how urban residents can be protected from **external network disruptions**, and how **metabolically sovereign urban units** can be constructed.